PrimeTime Suite Tool Invocation Commands

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PrimeTime Suite Tool Invocation Commands

This document describes the tool invocation commands supported by the PrimeTime Suite tool.

primetime

Runs the PrimeTime graphical user interface (GUI).

Syntax

primetime

```
-root_path path_name
[-display display_env_value]
[-file script_file]
[-no_init]
[-version]
[-multi_scenario]
```

Data Types

path_name	string
display_env_value	string
script_file	string

Arguments

`-root_path path_name`

Specifies the Synopsys root path.

`-display display_env_value`

Sets the DISPLAY environment variable to *display_env_value* for PrimeTime GUI if invoked from the pt_shell.

`-file script_file`

Executes the commands in the specified script file before displaying the initial pt_shell prompt. If the last statement in the script file is the *quit* or *exit* command, no prompt is displayed, and the command shell exits.

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`-no_init`

Prevents the execution any .synopsys_pt.setup startup files.

`-version`

Displays the version number and build date information and then exits.

`-multi_scenario`

Invokes PrimeTime in the master of the distributed processing mode. PrimeTime starts up and displays the pt_shell prompt just as in a single scenario session. A distributed multi-scenario analysis (DMSA) is carried out by one master PrimeTime process and multiple PrimeTime slave processes.

`-constraints`

Invokes PrimeTime in constraints consistency mode. PrimeTime starts up and displays a ptc_shell prompt. This mode opens a GUI, so the DISPLAY environment variable must point to a valid graphical environment.

`-output_log_file` *`log_file`*

Writes the tool session output to both the specified file and the terminal. This is equivalent to using the *tee* command in the host environment, except that the startup banner and tool exit message are not included in the output file.

`-container`

Invokes PrimeTime in a container. You can also set the SNPS_CONTAINER environment variable to 1 before invoking the tool. The Synopsys Container environment must be installed and configured.

Description

This command invokes the graphical user interface (GUI) to pt_shell. It executes all pt_shell startup files, commands, and command scripts. It also provides a graphical console for invoking the pt_shell commands and windows for graphical visualization of timing information.

The Main window is the graphical counterpart to pt_shell and is the only window that appears initially. The Main window provides a graphical console for executing pt_shell commands and viewing the results. Commonly used commands are also accessible from the Main window's menus. Lastly, the graphical visualization windows are invoked from the Main window.

The Histogram window can display a histogram of Endpoint Slack, Path Slack, Net Capacitance, or Design Rule Cost information for the current design. The histograms give an overall view of the state of the design. The Path Profiler window can be used to examine profiles of specific paths and nets identified in the Histogram window.

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The Path Schematic Window provides a schematic view of the path specified from the Histogram Window. In addition, pin names with rise/fall arrival value, net capacitance and cell names are annotated.

The Report window organize the reports generated and provides simple viewing and search capabilities to browse the reports.

See Also

- pt_shell

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pt_shell

Runs the PrimeTime command shell.

Syntax

pt_shell

```
-root_path path_name
[-gui]
[-display display_env_value]
[-file script_file]
[-no_init]
[-x command]
[-version]
[-multi_scenario]
[-output_log_file file_name]
[-adv_constraints]
[-constraints]
[-container]
```

Data Types

path_name	string
display_env_value	string
script_file	string
command	string
file_name	string

Arguments

`-root_path` *path_name*

Specifies the Synopsys root path.

`-gui`

Invokes the PrimeTime GUI immediately upon startup.

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`-display` *`display_env_value`*

Sets the DISPLAY environment variable to *display_env_value* for PrimeTime GUI if invoked from the *pt_shell* command.

`-file` *`script_file`*

Executes the commands in the specified script file before displaying the initial pt_shell prompt. If the last statement in the script file is the *quit* or *exit* command, no prompt is displayed, and the command shell exits.

`-x` *`command`*

Executes the pt_shell statements in *command* before displaying the initial pt_shell prompt. You can enter multiple statements, separating each statement with a semicolon, with quotation marks around the entire set of command statements after the *-x* option. If you also use the *-file* option, the statements from the *-x* option are executed before the script from the *-file* option is executed. If the last statement entered is *quit*, no prompt is displayed and the command shell is exited.

`-no_init`

Directs the pt_shell not to execute any .synopsys_pt.setup startup files.

`-version`

Displays the version number and build date information, and then exits.

`-multi_scenario`

Invokes PrimeTime in the master of the distributed processing mode. PrimeTime starts up and displays the pt_shell prompt just as in a single scenario session. A distributed multi-scenario analysis (DMSA) is carried out by one manager PrimeTime process and multiple PrimeTime worker processes.

`-adv_constraints`

Invokes PrimeTime in the advanced constraints mode.

`-constraints`

Invokes PrimeTime in constraints consistency mode. PrimeTime starts up and displays a ptc_shell prompt. This mode opens a GUI, so the DISPLAY environment variable must point to a valid graphical environment.

`-output_log_file` *`log_file`*

Writes the tool session output to both the specified file and the terminal. This is equivalent to using the *tee* command in the host environment, except that the startup banner and tool exit message are not included in the output file.

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`-container`

Invokes PrimeTime in a container. You can also set the SNPS_CONTAINER environment variable to 1 before invoking the tool. The Synopsys Container environment must be installed and configured.

Description

Interprets and executes PrimeTime timing analysis commands. The pt_shell command interpreter is based on Tcl (Tool Command Language).

The pt_shell executes commands until you terminate the session with the *quit* or *exit* command. During interactive mode, you can also terminate the pt_shell session by using Ctrl+D.

To cancel or interrupt the command currently executing in the pt_shell, type Ctrl+C. The time it takes for a command to process an interrupt (stop what it is doing and continue with the next command) depends upon the size of the design and the type of command. If you enter Ctrl+C three times before a command responds to the interrupt, pt_shell exits, and the following message is displayed:

```
Information: Process terminated by interrupt.
```

See Also

- primetime

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start_multi_user_client

Script to start a Multi User client session.

Syntax

start_multi_user_client

```
-session_id session_id
[-file script_file]
[-output output_log]
[-display display_env_value]
[-queue]
```

Data Types

session_id	string
script_file	string
output_log	string
display	string

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Arguments

`-session_id` *`session_id`*

Specifies the session ID to identify and connect to Multi User Server session. It is a unique string displayed at the Server session while starting Multi User mode which helps identify the port on which the Server is waiting for connection requests.

`-display` *`display_env_value`*

Specifies the display port where the GUI window will be started to run client in interactive mode. The value is picked from *DISPLAY* environment variable if this option is not specified.

`-file` *`script_file`*

Executes the commands in the specified script file in non interactive mode. The script is executed parallel to other clients and output is written to the file specified with *-output* option.

`-output` *`log_file`*

Writes the client session output to the specified file. It is supported with both interactive (GUI) and non-interactive mode. By default, a ${client_name}.out file will be created in the client working directory which will contain the output logs.

`-queue`

Allows queuing of client if request is sent after the maximum client limit of the Multi User session has been reached. By default, queueing is disabled and an appropriate error message will be printed by start_multi_user_client command. This option will help override this behavior and the queued client will be started when a running client has completed execution.

Description

Initiates communication with Multi User Server and requests Read-only access to PrimeTime design to execute timing analysis commands in Multi User client. The client will only be allowed to call a set of reporting commands that do not change the timing state of the design. It will have access to read all PrimeTime variables and will be allowed to update any variable that do not cause any change to the computed timing values. If the client user calls any command or updates any variable that changes timing state and triggers an incremental timing update, a severe error will be reported and the client will be terminated immediately.

The Multi User feature supports execution of client commands in two modes: interactive (GUI) and non-interactive.

In interactive mode, the script launches a GUI instance of the server PrimeTime design at client's display. The display configuration where the GUI is being launched will be printed

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by the *start_multi_user_client* script. Users can change the value of display port by using the *-display* option. The GUI window will contain a pt_shell prompt where the client users can input commands and a console where the output of client commands will be printed.

In non-interactive mode, the client user can write a script file containing list of all the PrimeTime read-only commands it wants to execute and pass the name of the script file with *-file* option. Users must explicitly provide path of an output file where the result of the client commands needs to be written with *-output* option. An appropriate error message will be printed if user does not provide this value.

The script is non blocking and returns control to UNIX shell after communicating client request to the Multi User Server.

Before calling this script, the client user must meet the following requirements:

Must be logged in to the same host as server.

User must be part of the UNIX group specified while starting Multi User session.

Must ensure the client's working directory is under access group ownership and group bits for read-write access is set properly.

From the above, #1 ensures that security of Multi User server design is imposed at system level by restricting unauthorized users from logging into the host. #2 and #3 ensures that client user has Read-Write access to output files written by client subprocess. Failure to meet any of the above requirements will cause the script to fail. An appropriate error message will be printed indicating the reason of failure and steps to resolve it.

Examples

Usage and output of *start_multi_user_client* script is as follows:

```
start_multi_user_client -session_id 32:5120
  Information: Setting 'hostname:1234' as display for Multi-User client.
 (MC-141)
  Information: Client logs will be available at
 '/path/client123.out' (MC-144)
  Information: Starting Multi-User client. (MC-152)
```

See Also

- start_multi_user_server
- report_multi_user_server
- stop_multi_user_server
- stop_multi_user_client